TRIBHUVAN UNIVERSITY

Institute of Science and Technology

A CASE STUDY REPORT

ON

ADOPTION OF CLOUD COMPUTING IN NEPAL (GOVERNMENT AND PRIVATE SECTORS)

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A CASE STUDY REPORT SUBMITTED TO THE DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE DEGREE OF BACHELOR IN SCIENCE IN COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

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10th September 2026

EXECUTIVE SUMMARY

This case study report examines the adoption trajectory of cloud computing across Nepal's government and private sectors, situating the analysis within the country's evolving legal, financial, and technical landscape. Nepal's public sector cloud footprint is anchored by the Government Integrated Data Center at Singha Durbar and its Disaster Recovery Center in Hetauda, both operated under the National Information Technology Center through the Integrated Data Management Center. These facilities host the domains, email systems, and mission critical applications of more than two thousand five hundred government offices, including all seven hundred fifty-three local units, along with data belonging to Nepal Rastra Bank, the Home Ministry, Nepal Police, and the Office of the Company Registrar. This infrastructure continues to rely on comparatively conventional virtualization rather than fully elastic, self service cloud provisioning, and its resilience was tested directly during the unrest of September 2025 in Kathmandu, when fire damage near Singha Durbar threatened the primary data center while the Hetauda facility preserved continuity of essential digital services; this episode has become a reference case for disaster recovery planning throughout the report.

On the regulatory front, the Data Center and Cloud Service (Operation and Management) Directive 2081, issued by the Ministry of Communication and Information Technology on 10 February 2025 under Section 79 of the Electronic Transactions Act 2063, has for the first time formally required data centers and cloud service providers operating in Nepal, whether government, public, or private, to register with the Department of Information Technology and to satisfy defined security, tier, and business continuity criteria. Ncell became the first company listed under this regime by virtue of its Tier 3 data center in Nakkhu, Lalitpur, an early institutional milestone that this report treats as indicative of the market's current stage of maturity.

Private sector adoption is markedly bifurcated. Banking, financial services, and insurance institutions operate under Nepal Rastra Bank's Information Technology Guidelines, which since 2012 have permitted regulated offshore data hosting through cloud computing while still compelling banks to maintain domestic disaster recovery capacity; in practice, most banks retain their core banking systems on premises or in domestic colocation and use foreign cloud services selectively for peripheral workloads. Nepal's software export and information technology enabled services industry, by contrast, is heavily reliant on Amazon Web Services, Microsoft Azure, and Google Cloud Platform for globally aligned software delivery, a dependence complicated by Nepal Rastra Bank's foreign exchange regime. Ordinary citizens and small enterprises have historically been capped at five hundred United States dollars a year on prepaid dollar cards used to settle foreign cloud invoices; a unified circular amendment reported in April 2026 introduced dedicated allowances of three thousand to five thousand United States dollars a year for information technology, communication, and broadcasting businesses purchasing software and cloud services, and larger allowances for exporters of

digital services, a change that materially affects the total cost of ownership calculus explored in Chapter 7.

The report finds that domestic data centers, including those operated by Ncell, Nepal Telecom, and other listed or emerging providers, offer stronger data sovereignty guarantees and lower latency for Nepal facing workloads, but presently lag global hyperscalers in managed platform services, elasticity, and geographic redundancy. The central conclusion is that Nepal's cloud future depends on harmonizing the newly enacted Directive 2081 licensing regime with continued foreign exchange liberalization for technology purchases, sustained investment in the Integrated Data Management Center's automation and security operations capability, and a coordinated skills pipeline linking academia with internationally recognized cloud certification standards.

TABLE OF CONTENTS

EXECUTIVE SUMMARY ii
TABLE OF CONTENTS iv
LIST OF TABLES vi
CHAPTER 1: INTRODUCTION 1
1.1 Theoretical and Architectural Foundations of Cloud Computing 1
1.2 The Cloud Paradigm Shift in Developing Economies 1
1.3 Scope of the Report 2
1.4 Methodology 2
1.5 Research Questions and Problem Statement 2
CHAPTER 2: CLOUD TAXONOMY AND ARCHITECTURAL MODELS 4
2.1 Service Models 4
2.2 Deployment Models 4
2.3 Shared Responsibility Model 4
CHAPTER 3: OBJECTIVES OF THE STUDY 6
CHAPTER 4: POLICY, LEGAL, AND REGULATORY FRAMEWORK IN NEPAL 7
4.1 The Data Center and Cloud Service (Operation and Management) Directive 2081 7
4.2 Data Sovereignty, Privacy, and National Security Instruments 8
4.3 Financial Regulations: Nepal Rastra Bank and Foreign Exchange Constraints 8
4.4 Digital Nepal Framework 2019 9
CHAPTER 5: PUBLIC SECTOR ADOPTION: THE NEPAL GOVERNMENT CLOUD (G-CLOUD) 11
5.1 Infrastructure Backbone: GIDC, DRC, and Institutional Ownership 11
5.2 Technical Architecture of G Cloud 11
5.3 Workload Migration: Case Reviews of Major Public Systems 11
5.4 Operational Deficits and the September 2025 Stress Test 12
CHAPTER 6: PRIVATE SECTOR CLOUD ADOPTION LANDSCAPE 14
6.1 Banking, Financial Services, and Insurance (BFSI) 14
6.2 Software Outsourcing and Technology Export Sector 14
6.3 E Commerce and Digital Payment Rails 14
6.4 Small and Medium Enterprises (SMEs) and Startups 15
6.5 Domestic Commercial Data Centers and Local Cloud Infrastructure 15
CHAPTER 7: COMPARATIVE ANALYSIS: DOMESTIC INFRASTRUCTURE VERSUS GLOBAL HYPERSCALERS 17
7.1 Performance and Latency 17
7.2 Total Cost of Ownership (TCO) and Economics 17

7.3 Financial Friction and Foreign Exchange Billing 17
7.4 Feature Parity and Ecosystem Maturity 18
CHAPTER 8: RISKS, CYBERSECURITY, AND OPERATIONAL CHALLENGES 20
8.1 Data Sovereignty, Extraterritorial Jurisdiction, and Vendor Lock In 20
8.2 Cyber Threat Landscape 20
8.3 Public Sector Institutional Bottlenecks 20
8.4 International Network Transit Vulnerability 21
CHAPTER 9: EMERGING TRENDS AND FUTURE OUTLOOK 22
CHAPTER 10: CONCLUSION AND STRATEGIC SYNTHESIS 23
CHAPTER 11: ACTIONABLE RECOMMENDATIONS 24
11.1 Government Policy Makers and the Ministry of Communication and Information Technology 24
11.2 Nepal Rastra Bank and Financial Regulators 24
11.3 Public Cloud Infrastructure Operators (NITC and the Integrated Data Management Center) 24
11.4 Private Sector and Commercial Data Center Operators 24
11.5 Academia and Human Capital Development 25
REFERENCES 26

LIST OF TABLES

Table 1: Comprehensive Classification Framework of Cloud Computing Deployment in Nepal

Table 2: Legal and Regulatory Matrix Governing Cloud Computing and Data Infrastructure in Nepal

Table 3: Technical Specifications, Capacity Metrics, and Operational Benchmark of Government Cloud Infrastructure

Table 4: Sectoral Cloud Adoption Matrix in Nepal's Private Industry (BFSI, Technology Export, Fintech, SMEs)

Table 5: Major Domestic Commercial Data Centers and Local Cloud Infrastructure Providers in Nepal

Table 6: Comparative Benchmark of Domestic Cloud and Colocation Providers versus International Hyperscalers

Table 7: Indicative Total Cost of Ownership Multi Year Model for Nepali Enterprises

Table 8: Risk Categorization, Operational Impact, and Mitigation Matrix for Cloud Deployments in Nepal

Table 9: Stakeholder Action Matrix with Prioritized Implementation Horizons

CHAPTER 1: INTRODUCTION

1.1 Theoretical and Architectural Foundations of Cloud Computing

Cloud computing is defined by the United States National Institute of Standards and Technology as a model for enabling ubiquitous, convenient, on demand network access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort or service provider interaction. This definition identifies five essential characteristics: on demand self service, broad network access, resource pooling, rapid elasticity, and measured service. These characteristics distinguish cloud computing from traditional data center hosting, in which capacity is fixed at the time of procurement and cannot be elastically expanded or contracted in response to demand.

Three canonical service models structure the cloud market. Infrastructure as a Service (IaaS) provides virtualized compute, storage, and networking on which the tenant installs its own operating system and application stack. Platform as a Service (PaaS) abstracts away infrastructure management and exposes a managed runtime, database, or application platform. Software as a Service (SaaS) delivers a complete, provider managed application accessed over the network, typically through a browser or thin client. A fourth, increasingly significant pattern is serverless computing, in which the provider dynamically manages the allocation of machine resources and the tenant is billed only for the actual computation consumed, often orchestrated through containerized microservices built on Docker and coordinated through Kubernetes.

Four deployment models complete the taxonomy: public cloud, in which infrastructure is provisioned for open use by the general public and owned by a cloud service provider; private cloud, provisioned exclusively for a single organization; community cloud, shared among organizations with common concerns; and hybrid cloud, a composition of two or more distinct infrastructures bound together by standardized technology enabling data and application portability. Nepal's cloud landscape, as this report demonstrates, is best understood as a hybrid and multi cloud environment in transition, where a sovereign public sector cloud, several domestic private clouds, and extensive consumption of international public hyperscaler services coexist under a single, newly formalized regulatory umbrella.

1.2 The Cloud Paradigm Shift in Developing Economies

For capital constrained, landlocked economies such as Nepal, cloud computing represents a structural shift from capital expenditure (CapEx) intensive information technology procurement toward an operational expenditure (OpEx) consumption model. Traditional data center ownership in Nepal has historically required importing physical servers, storage arrays, and networking equipment through customs procedures that add cost, lead time, and foreign exchange exposure, followed by ongoing burdens of power provisioning, cooling, and hardware refresh cycles in a market with limited access to enterprise grade replacement parts.

The cloud model substitutes this fixed cost structure with variable, usage based billing, offering agility for a private sector that must scale rapidly to compete internationally and offering the public sector a theoretical path to modernizing legacy systems without large upfront capital allocations that are difficult to secure through the annual government budgeting cycle. However, this same shift transforms domestic capital expenditure into a foreign currency operational liability when the provider is an international hyperscaler, a tension explored at length in Chapters 6 and 7.

1.3 Scope of the Report

This report undertakes an exhaustive analysis of cloud computing adoption across five interlocking domains of the Nepali economy: the central government and its constituent ministries and departments; the banking, financial services, and insurance sector regulated by Nepal Rastra Bank; the domestic information technology export and outsourcing industry; telecommunications operators that double as commercial data center and cloud service providers; and small and medium enterprises together with early stage startups. The analysis integrates statutory instruments, institutional case histories, and comparative technical benchmarking to produce a holistic account of where Nepal presently stands in its cloud adoption journey and what structural barriers remain.

1.4 Methodology

The report is constructed through desk based empirical research combining primary legal instruments, including the Data Center and Cloud Service (Operation and Management) Directive 2081, the Electronic Transactions Act 2063, the Individual Privacy Act 2075, the National Cyber Security Policy 2080, and Nepal Rastra Bank's Information Technology Guidelines, with secondary sources comprising government agency disclosures, telecommunications regulatory filings, industry client alerts published by Nepali law firms, financial press coverage, and institutional statements from the National Information Technology Center and the Integrated Data Management Center. Where precise operational figures are not publicly disclosed by domestic providers, the report clearly labels its estimates as indicative and illustrative rather than presenting them as verified statistics, in keeping with responsible academic practice for a rapidly evolving and only partially documented market.

1.5 Research Questions and Problem Statement

The report is organized around the following research questions: first, to what extent has Nepal's public sector transitioned from conventional data center hosting to genuinely elastic cloud provisioning, and what operational deficits constrain that transition; second, how do Nepal Rastra Bank's financial regulations and the broader foreign exchange regime shape private sector cloud purchasing decisions; third, what data sovereignty and cyber security obligations now apply to data centers and cloud service providers under Directive 2081 and related instruments; fourth, how does the total cost of ownership and feature parity of domestic

private cloud infrastructure compare with international hyperscaler alternatives; and fifth, what concrete, differentiated actions can policy makers, regulators, infrastructure operators, and academic institutions take to accelerate secure and sustainable cloud adoption. The underlying problem statement is that Nepal, as a landlocked nation with limited submarine cable access, constrained foreign exchange reserves for digital services, and an emergent rather than mature regulatory architecture, faces a distinctive set of adoption barriers that differ materially from those documented in coastal, higher income economies, warranting a dedicated case study rather than the uncritical importation of global best practice frameworks.

CHAPTER 2: CLOUD TAXONOMY AND ARCHITECTURAL MODELS

2.1 Service Models

Within Nepal, all three canonical service models are represented, though unevenly. Infrastructure as a Service is the dominant model both for the Government Integrated Data Center, which primarily leases virtual machines and storage to line ministries, and for the software export industry, which provisions IaaS compute directly from Amazon Web Services, Microsoft Azure, and Google Cloud Platform. Platform as a Service adoption remains comparatively shallow domestically; while Nepali developers make extensive use of managed platform offerings from foreign hyperscalers such as managed databases, container orchestration services, and serverless function platforms, no domestic provider yet offers a mature, general purpose PaaS layer. Software as a Service adoption is widespread at the consumption level, particularly Google Workspace and Microsoft 365 for productivity, and various fintech and accounting SaaS platforms for small business operations, but Nepal has yet to produce SaaS products at meaningful export scale. Containerization through Docker and orchestration through Kubernetes are increasingly standard practice among Nepal's technology export firms serving international clients, though public sector adoption of container orchestration remains at an early, largely pilot stage.

2.2 Deployment Models

Nepal's cloud environment exhibits four distinguishable deployment patterns operating in parallel. Private on premises infrastructure remains the default for core banking systems and for many government line agencies that have not yet migrated workloads to the Government Integrated Data Center. Hosted private cloud, delivered through domestic colocation and data center operators such as Ncell, is increasingly used by banks and larger enterprises seeking a middle path between full ownership and full reliance on foreign public cloud. A nascent sovereign public cloud exists in the form of the Government Integrated Data Center's shared virtualization platform, which multiple ministries draw upon as tenants, though it does not yet offer the self service provisioning portal, granular metering, or elastic auto scaling associated with mature public cloud platforms. Finally, hybrid and multi cloud topologies, combining domestic private infrastructure with one or more foreign hyperscaler accounts, are the emerging norm for technology export firms, digital payment switches, and larger private enterprises that must balance data sovereignty obligations against the operational maturity of international platforms.

2.3 Shared Responsibility Model

Across all deployment models operating in Nepal, security and operational responsibility is divided between the tenant organization and the infrastructure or service provider according to

the shared responsibility model. Under IaaS, the provider is responsible for the physical facility, network, and virtualization layer, while the tenant remains responsible for the guest operating system, middleware, application code, and data classification. Under PaaS, the provider additionally assumes responsibility for the runtime and middleware, leaving the tenant accountable chiefly for application logic and data. Under SaaS, the provider is responsible for nearly the entire stack, with the tenant retaining responsibility principally for user access management and data governance. This division carries particular significance in Nepal because Directive 2081 and Nepal Rastra Bank's guidelines impose compliance obligations, such as data classification, encryption, and business continuity planning, that in practice fall on the tenant organization regardless of which party technically operates the underlying infrastructure, a distinction many smaller Nepali organizations have not yet fully internalized.

Table 1: Comprehensive Classification Framework of Cloud Computing Deployment in Nepal

Dimension	Categories Present in Nepal	Representative Examples
Service Model	IaaS (dominant); PaaS (foreign hyperscaler only); SaaS (widespread consumption); Serverless and containerized workloads (export sector)	G Cloud virtual machines; AWS RDS and Azure App Service; Google Workspace, Microsoft 365; Docker and Kubernetes based export pipelines
Deployment Model	Private on premises; hosted private and colocation; sovereign public (nascent); hybrid multi cloud	Bank core banking data centers; Ncell Tier 3 facility; NITC G Cloud; fintech switches spanning domestic and foreign platforms
Primary Tenant Category	Central government ministries and departments; BFSI institutions; technology export firms; telecom operators; SMEs and startups	IRD, DoP, PSC; commercial banks; software outsourcing companies; Ncell and Nepal Telecom; small e commerce ventures
Governing Instrument	Directive 2081; NRB IT Guidelines; sector specific circulars	Data Center and Cloud Service (Operation and Management) Directive 2081; NRB Unified Circulars

CHAPTER 3: OBJECTIVES OF THE STUDY

The primary objectives of this case study report are as follows:

1. To examine the current architectural and operational state of cloud and data center infrastructure in Nepal's public and private sectors.

2. To analyze the legal, regulatory, and financial framework governing data centers and cloud service providers, with particular attention to the Data Center and Cloud Service (Operation and Management) Directive 2081 and Nepal Rastra Bank's foreign exchange and information technology guidelines.

3. To evaluate the technical maturity of the Government Integrated Data Center and its associated Disaster Recovery Center in Hetauda relative to internationally recognized cloud operating standards.

4. To compare the cost, performance, and feature parity of domestic cloud and colocation providers against global hyperscalers serving the Nepali market.

5. To identify the principal cyber security, data sovereignty, and institutional risks associated with cloud adoption in Nepal.

6. To propose differentiated, actionable recommendations for government policy makers, financial regulators, infrastructure operators, private enterprise, and academia.

The secondary objectives of the study are as follows:

1. To document institutional case histories, including the migration of major public digital services onto the Government Integrated Data Center and the operational lessons drawn from the September 2025 disruption in Kathmandu.

2. To construct a reference matrix of domestic commercial data centers and cloud providers for use by future researchers, policy analysts, and enterprise decision makers.

CHAPTER 4: POLICY, LEGAL, AND REGULATORY FRAMEWORK IN NEPAL

4.1 The Data Center and Cloud Service (Operation and Management) Directive 2081

The single most consequential regulatory development in Nepal's cloud landscape is the Data Center and Cloud Service (Operation and Management) Directive 2081, issued by the Ministry of Communication and Information Technology on 10 February 2025 (2081.10.15 in the Nepali calendar) under the authority of Section 79 of the Electronic Transactions Act 2063. Its preamble states that it is expedient to encourage the secure and confidential storage of data created by government, public, and private sectors within the country, and to make information technology systems reliable, secure, and effective through the listing and management of data centers and cloud service providers. The Directive requires every data center and cloud service provider, whether serving government, public, or private clients, to register separately with the Department of Information Technology before commencing or continuing operations, with existing operators required to apply within six months of the Directive's issuance, that is, by 31 July 2025. Applicants must submit documentation including a company or firm registration certificate, a security and privacy policy, a business continuity plan, a location map, technical personnel details, and Internet Protocol pool information; data center applicants must additionally furnish fire safety assurance, a building completion certificate, tier classification details, physical security methodology, and high level electrical design documentation, while cloud service applicants must submit agreements with their underlying data center and internet service or network service provider. The Department may delist a provider for non compliance, data misuse, dissolution, or voluntary withdrawal, and oversight of the regime rests with a Directorate Committee chaired by the Secretary of the Ministry of Communication and Information Technology, with the Directive's overall effectiveness scheduled for review two years after enforcement.

A particularly notable governance feature of the Directive is that Nepal's security agencies are required, subject to a grandfathering provision for infrastructure already collocated at the time of enforcement, to use only the data center and cloud services operated by the Integrated Data Management Center, reinforcing state control over the most sensitive government workloads even as private commercial listing expands elsewhere in the market. In April 2025, Ncell became the first company listed as both a data center and a cloud service provider under the new regime, following direct inspection of its Tier 3 facility in Nakkhu, Lalitpur, by the Department of Information Technology, an early but significant signal of the domestic private sector's capacity to meet the Directive's operational and security standards.

4.2 Data Sovereignty, Privacy, and National Security Instruments

The Electronic Transactions Act 2063 provides the foundational legal recognition for electronic records and digital signatures in Nepal and, as noted above, is the parent statute under which Directive 2081 was promulgated. The Individual Privacy Act 2075 establishes rights and obligations concerning the collection, storage, and processing of personal information by public and private bodies, creating an overlapping compliance obligation for any organization storing personal data in domestic or foreign cloud infrastructure. The National Cyber Security Policy 2080, endorsed by the Government of Nepal, articulates a national vision of building a secure and resilient cyberspace, names protection of critical national infrastructure, including government data hubs and financial systems, as an explicit policy objective, and calls for institutional structures such as a National Cyber Security Center and a national incident response capability, while candidly acknowledging weak legal and institutional structures and a shortage of skilled cyber security human resources as the principal implementation gaps.

4.3 Financial Regulations: Nepal Rastra Bank and Foreign Exchange Constraints

Nepal Rastra Bank's Information Technology Guidelines, first issued in 2012, require every commercial bank operating in the country to formulate a formal information technology security policy, maintain a documented business continuity and disaster recovery plan, and, notably, explicitly legalize the parking of bank data in foreign jurisdictions through cloud computing, provided the bank complies with the guideline's broader risk management, access control, and audit requirements within a two year compliance window. In practice, this provision has been interpreted conservatively, with most banks retaining core banking system data domestically or in supervised colocation while using foreign cloud services selectively for non core functions.

Nepal's foreign exchange regime, administered by Nepal Rastra Bank, has historically constrained how effectively private individuals and small enterprises can pay foreign cloud vendors. Since a March 2021 circular, commercial and development banks have issued prepaid dollar cards capped at a cumulative five hundred United States dollars per year per individual, usable only for online payments for goods and services and expressly barred from cash withdrawal or capital account transactions; this cumulative limit applies across all cards held by an individual regardless of how many banks issue them. This ceiling proved restrictive for freelancers, startups, and information technology firms whose recurring hyperscaler bills routinely exceed it. A Unified Circular amendment reported in April 2026 addressed this constraint directly for the technology sector: information technology, communication, and broadcasting businesses became eligible for dedicated prepaid cards of up to three thousand United States dollars annually for software and cloud purchases, rising to approximately five thousand United States dollars annually for entities earning foreign currency through service exports, with larger industry level allowances of up to one hundred thousand United States

dollars annually reported for software and technology procurement and up to twenty five thousand United States dollars annually for individual online service exporters, while the base individual personal limit of five hundred United States dollars per year was reported to remain in place for users without foreign income. Because this amendment postdates most secondary literature on Nepal's cloud economics, this report treats the higher figures as recently reported and recommends that practitioners verify current limits directly with Nepal Rastra Bank or their bank prior to relying on them for procurement planning.

4.4 Digital Nepal Framework 2019

The Digital Nepal Framework, released in 2019 and updated periodically thereafter, identifies eight sectors for digitally enabled transformation: digital foundation, agriculture, health, education, energy, tourism, finance, and urban infrastructure, structured around eighty distinct digital initiatives. The Digital Foundation sector, treated as the base layer underpinning all others, rests on three pillars of digital connectivity, digital skills, and digital governance, and includes initiatives directly relevant to this report such as expansion of the National Optical Fiber Network, establishment of a National Cyber Security Center, provincial data center establishment, and the pursuit of high speed internet connectivity for efficient delivery of public services. The Framework's own literature acknowledges that its success depends on public private partnership, given that private industry has led much of Nepal's technological transformation to date, and explicitly calls out data protection, security, and privacy as pillars for building public trust in digital government services, directly reinforcing the objectives later codified in Directive 2081 and the National Cyber Security Policy 2080.

Table 2: Legal and Regulatory Matrix Governing Cloud Computing and Data Infrastructure in Nepal

Instrument	Issuing Authority	Core Provision Relevant to Cloud Computing
Electronic Transactions Act 2063 (2008)	Government of Nepal (Parliament)	Legal recognition of electronic records and digital signatures; Section 79 empowers MoCIT to issue data center and cloud directives
Data Center and Cloud Service (Operation and Management) Directive 2081 (2025)	Ministry of Communication and Information Technology	Mandatory registration of data centers and cloud providers with the Department of Information Technology; tier, security, and continuity standards
Individual Privacy Act 2075	Government of Nepal	Rights and obligations concerning collection, storage, and processing of personal information
National Cyber Security Policy 2080	Government of Nepal	National vision for secure cyberspace; protection of critical infrastructure including data hubs and financial systems

Instrument	Issuing Authority	Core Provision Relevant to Cloud Computing
Nepal Rastra Bank Information Technology Guidelines (2012, as updated)	Nepal Rastra Bank	Mandatory IT security policy and disaster recovery plans for banks; conditional legalization of offshore cloud data parking
NRB Unified Circular on Foreign Exchange (as amended, reported April 2026)	Nepal Rastra Bank	Prepaid dollar card limits for individuals and IT sector entities used to pay foreign cloud vendors
Digital Nepal Framework 2019	Ministry of Communication and Information Technology	National digital transformation roadmap; Digital Foundation pillar covering connectivity, skills, and governance
Public Procurement Act 2063	Government of Nepal	Governs public sector procurement of infrastructure and cloud services, historically favoring lowest bidder evaluation

CHAPTER 5: PUBLIC SECTOR ADOPTION: THE NEPAL GOVERNMENT CLOUD (G-CLOUD)

5.1 Infrastructure Backbone: GIDC, DRC, and Institutional Ownership

Nepal's public sector cloud infrastructure is anchored by the Government Integrated Data Center at Singha Durbar, Kathmandu, originally established in 2009 as the Integrated Data Center with financial and technical support from the Government of the Republic of Korea through the Korea International Cooperation Agency (KOICA). A companion Disaster Recovery Center was subsequently built in Huprachaur, Hetauda, Makawanpur, becoming operational in 2019 to provide backup infrastructure and business continuity for the primary facility, with a further Korean government supported capacity enhancement project following in 2021. Both facilities are operated by the National Information Technology Center through its Integrated Data Management Center, which functions as the technical focal point for Nepal's electronic government implementation, handling domain registration, website hosting, network management, virtualization, and consultancy support for government bodies. As of the most recent public reporting, the center operates the websites and underlying infrastructure of more than two thousand five hundred government offices, including all seven hundred fifty three local government units, and stores data belonging to Nepal Rastra Bank, the Ministry of Home Affairs, Nepal Police, and the Office of the Company Registrar, among other agencies.

5.2 Technical Architecture of G Cloud

The Government Integrated Data Center's technical architecture follows a conventional virtualized data center model rather than a fully self service, elastically scaling public cloud platform. Compute virtualization is understood to rely on a mix of enterprise hypervisor platforms consistent with regional government cloud deployments, commonly reported in industry literature as including VMware vSphere alongside open source alternatives such as OpenStack and KVM for selected workloads, supporting server consolidation and centralized management of virtual machines allocated to tenant ministries. Storage is provisioned through a combination of storage area network (SAN) and network attached storage (NAS) systems to support the availability and backup requirements of hosted government applications, while networking is understood to follow a software defined approach for internal segmentation of ministry tenant environments within the shared Singha Durbar facility. Because NITC does not publish detailed technical architecture disclosures for security reasons, the specific configuration details in this section should be read as representative of standard government private cloud practice in South Asia rather than as officially confirmed specifications.

5.3 Workload Migration: Case Reviews of Major Public Systems

A substantial share of Nepal's digital government services now depends, directly or indirectly, on the Government Integrated Data Center's infrastructure. The backend supporting the

Nagarik App, Nepal's flagship citizen facing digital services application, along with systems used by the Department of Passports for passport issuance workflows, the Public Service Commission for civil service examination administration, land records management systems used by the Department of Land Management and Archive, and the Inland Revenue Department's Integrated Tax System for tax administration, all rely in whole or in part on shared government data center infrastructure, whether hosted directly at Singha Durbar or through agency specific servers collocated within the facility. This concentration of critical citizen services onto a single physical and administrative infrastructure amplifies both the efficiency gains from shared virtualization and the systemic risk exposure discussed further in Chapter 8.

5.4 Operational Deficits and the September 2025 Stress Test

Several operational deficits constrain the Government Integrated Data Center's ability to function as a genuine cloud platform rather than a shared virtualized hosting facility. High concurrency public services, particularly during peak demand events such as examination result publication or annual tax filing deadlines, have historically experienced performance degradation, reflecting the absence of automated horizontal scaling capability. Hardware lifecycle management is complicated by Nepal's constrained procurement cycles and foreign exchange dependent import processes for enterprise hardware, contributing to component aging that current NITC leadership has publicly attributed to at least one documented data loss incident: in 2023, the NITC executive director confirmed that a fault in a server's storage subsystem led to a digital data loss event affecting an undetermined number of government bodies, with recovery efforts requiring international technical support, and with NITC officials acknowledging that backup coverage across all hosted agencies was, at the time, incomplete.

The most consequential real world test of this infrastructure's resilience occurred during the September 2025 unrest in Kathmandu, when fire and looting struck government buildings in and around Singha Durbar. According to the head of the Integrated Data Management Center, the primary data center and the Hetauda disaster recovery office both survived unharmed even as fire raged nearby, preserving continuity for the more than two thousand five hundred government office websites and associated citizen records the center hosts; officials stated publicly that had the data center itself been destroyed, the damage to government continuity would have far exceeded the physical destruction of buildings, since countless paper records lost in the unrest would otherwise have had no digital backup. This episode simultaneously validated the value of the Hetauda disaster recovery investment and exposed the concentration risk inherent in Nepal having, in effect, a single primary government data center campus. NITC has separately indicated plans to expand data center capacity to Khumaltar and Nepalgunj and, over the longer term, to establish data centers across all seven provinces, a trajectory examined further in Chapter 9.

Table 3: Technical Specifications, Capacity Metrics, and Operational Benchmark of Government Cloud Infrastructure

Parameter	Government Integrated Data Center (Singha Durbar)	Disaster Recovery Center (Hetauda)
Year Established	2009 (with KOICA support)	Operational since 2019, with further capacity support in 2021
Operating Authority	National Information Technology Center / Integrated Data Management Center	National Information Technology Center / Integrated Data Management Center
Primary Function	Hosting, domain registration, email, virtualization, and network management for government bodies	Business continuity and backup infrastructure for GIDC in the event of disaster
Approximate Tenant Scale	Websites and services for over 2,500 government offices, including all 753 local units	Backup capacity scoped to critical agency data as prioritized by NITC
Virtualization Approach (indicative)	Hypervisor based server virtualization with SAN/NAS storage	Mirrored or periodic replication architecture (indicative)
Documented Resilience Event	Not directly damaged during September 2025 Kathmandu unrest despite nearby fire	Maintained continuity throughout the September 2025 unrest
Known Operational Constraint	2023 storage subsystem fault caused a documented data loss incident; incomplete backup coverage across all hosted agencies acknowledged by NITC	Backup scope historically limited to agencies that specifically request it

CHAPTER 6: PRIVATE SECTOR CLOUD ADOPTION LANDSCAPE

6.1 Banking, Financial Services, and Insurance (BFSI)

Nepal's BFSI sector operates under the most tightly regulated cloud environment in the country. Nepal Rastra Bank's Information Technology Guidelines require every commercial bank to maintain a documented information technology security policy, implement layered network security including firewalls between internal and external networks and across geographically separate sites, encrypt sensitive data such as personal identification numbers and passwords both in transit and at rest, and maintain a business continuity and disaster recovery plan sufficient to protect customers from service disruption caused by events such as earthquakes. While the guidelines explicitly legalize the parking of bank data in foreign jurisdictions through cloud computing, most Nepali banks have adopted a conservative hybrid posture in practice: core banking systems and the most sensitive customer data typically remain on premises or in domestic colocation facilities such as those operated by Ncell or Nepal Telecom, while foreign cloud platforms are used selectively for lower sensitivity workloads such as marketing analytics, customer relationship management, or disaster recovery replication for non core systems. This hybrid posture reflects both regulatory caution and the practical reality that Nepal Rastra Bank's periodic onsite and offsite supervision continues to scrutinize IT risk management, access control, and outsourcing arrangements closely.

6.2 Software Outsourcing and Technology Export Sector

Nepal's information technology enabled services and software outsourcing industry represents the segment of the economy with the deepest and most sophisticated cloud adoption. Firms serving international clients in software product development, quality assurance, and managed services routinely provision infrastructure natively on Amazon Web Services, Microsoft Azure, or Google Cloud Platform, both because international clients frequently mandate a specific hyperscaler for contractual or compliance reasons and because these platforms offer continuous integration and continuous deployment (CI/CD) tooling, managed database services, and global points of presence that no domestic provider currently matches. Microservices architectures, containerized through Docker and orchestrated through Kubernetes, are standard practice among more mature Nepali technology exporters, enabling them to align delivery practices with global engineering teams and international quality certifications.

6.3 E Commerce and Digital Payment Rails

Nepal's digital payment infrastructure, including the interbank payment switch connectIPS and wallet based platforms such as eSewa and Khalti, occupies a particularly sensitive position in the cloud adoption landscape because these systems must simultaneously satisfy Payment Card

Industry Data Security Standard (PCI DSS) obligations, Nepal Rastra Bank oversight, and demanding local latency targets for real time transaction authorization. In practice, payment switches and wallet operators tend to favor in house private cloud or domestic colocation deployments for their core transaction processing systems, reserving hybrid use of foreign hyperscaler services for analytics, fraud detection modeling, and non transactional workloads where the latency and sovereignty constraints are less acute.

6.4 Small and Medium Enterprises (SMEs) and Startups

Nepal's SME and startup segment has undergone a visible transition over the past decade, moving away from shared Linux cPanel hosting, once the default for small business websites and basic applications, toward cost efficient cloud instances offered by providers such as DigitalOcean, Linode, and Amazon Web Services Lightsail, alongside widespread adoption of SaaS productivity and collaboration tools such as Google Workspace and Microsoft 365. This transition has been constrained less by technical unfamiliarity than by the financial and regulatory friction discussed in Section 4.3: until the 2026 amendment introducing higher IT sector allowances, many small operators found the five hundred United States dollar annual dollar card ceiling a binding constraint on how much foreign cloud infrastructure they could legally purchase, pushing some toward informal payment workarounds or toward domestic hosting alternatives regardless of feature parity considerations.

Table 4: Sectoral Cloud Adoption Matrix in Nepal's Private Industry (BFSI, Technology Export, Fintech, SMEs)

Sector	Dominant Deployment Pattern	Primary Regulatory Driver	Principal Adoption Constraint
Banking, Financial Services, and Insurance	Hybrid: on premises or domestic colocation for core banking; selective foreign cloud for peripheral workloads	NRB IT Guidelines; Directive 2081	Conservative regulatory posture; supervisory scrutiny of outsourcing
Software Export and ITeS	Native hyperscaler (AWS, Azure, GCP) IaaS and PaaS	Client contractual mandates; Directive 2081 registration for domestic components	Foreign exchange settlement; talent retention
E Commerce and Digital Payments	In house private cloud or domestic colocation for core switching; hybrid for analytics	PCI DSS; NRB oversight; Directive 2081	Latency sensitivity; sovereignty requirements
SMEs and Startups	Transitioning from shared hosting to VPS and SaaS	Consumer protection norms; informal practice	Foreign exchange card limits; digital literacy

6.5 Domestic Commercial Data Centers and Local Cloud Infrastructure

A small but growing cohort of domestic providers now offers commercial data center and, increasingly, cloud infrastructure services within Nepal. Telecommunications operators have been the most prominent entrants: Ncell operates a Tier 3 data center facility in Nakkhu,

Lalitpur, and became the first company in Nepal listed under Directive 2081 as both a data center and cloud service provider following inspection by the Department of Information Technology, positioning it to serve banks, government agencies, and large commercial organizations. Nepal Telecom similarly operates data center capacity leveraging its nationwide network infrastructure. A number of smaller commercial providers, often described in industry commentary as offering colocation, virtual private server, and emerging private cloud services, have also entered the market, though public documentation of their tier classification, capacity, and specific service catalogs remains limited; this report accordingly presents them descriptively rather than with fabricated precision, and recommends that specific technical claims about any individual provider be verified directly against that provider's current published specifications and its Directive 2081 listing status.

Table 5: Major Domestic Commercial Data Centers and Local Cloud Infrastructure Providers in Nepal

Provider	**Ownership Category**	**Reported Service Focus**	**Directive 2081 Listing Status (as of this report)**
Ncell (Nakkhu, Lalitpur)	Private telecommunications operator	Tier 3 colocation; data center and cloud services for banks, government, and enterprise	First company listed as both data center and cloud service provider
Nepal Telecom	State owned telecommunications operator	Data center and hosting services leveraging national network infrastructure	Subject to the same registration requirement as all providers under Directive 2081
Other domestic colocation and VPS providers (various, e.g. DataSpace, Silver Lining, Access World and similar entities named in industry commentary)	Private commercial operators	Colocation, virtual private servers, and emerging private cloud offerings for enterprise and SME clients	Registration status varies; readers should verify current listing directly with the Department of Information Technology
National Information Technology Center (Government Integrated Data Center)	Government (public sector)	Shared virtualization hosting for ministries and departments; not a commercial offering	Government facilities are subject to a distinct oversight track under Directive 2081

CHAPTER 7: COMPARATIVE ANALYSIS: DOMESTIC INFRASTRUCTURE VERSUS GLOBAL HYPERSCALERS

7.1 Performance and Latency

Because Nepal has no domestic hyperscaler availability zone, workloads hosted on Amazon Web Services, Microsoft Azure, or Google Cloud Platform are typically served from the nearest regional infrastructure, most commonly Mumbai (AWS ap south 1) or Hyderabad (AWS ap south 2) in India, or from Singapore, with international transit carried over terrestrial fiber links routed through India or, to a lesser extent, China. Round trip latency from Kathmandu to these regions is generally reported in industry practice to fall in the range of approximately twenty to sixty milliseconds for Indian regions and somewhat higher for Singapore, compared with single digit to low double digit millisecond latencies achievable for workloads hosted domestically within Kathmandu Valley colocation facilities. For latency sensitive applications such as real time payment switching or high frequency transaction processing, this differential favors domestic hosting; for latency tolerant workloads such as batch analytics, backup replication, or asynchronous business applications, the differential is of limited practical consequence, which is why Nepali enterprises frequently adopt a workload specific rather than blanket hosting strategy.

7.2 Total Cost of Ownership (TCO) and Economics

Domestic private infrastructure and colocation typically require higher upfront capital investment, denominated substantially in foreign currency for imported hardware, but generate predictable, largely Nepali rupee denominated ongoing costs once deployed. International hyperscaler consumption, conversely, requires minimal upfront capital but generates recurring, usage sensitive operational costs denominated in United States dollars, which are directly exposed to Nepal Rastra Bank's foreign exchange regime, to Nepali rupee depreciation against the dollar over time, and to withholding tax and value added tax treatment of cross border digital service payments. For a Nepali enterprise's total cost of ownership analysis, the deciding factors are therefore not merely price per compute unit, but foreign exchange settlement capacity, tax treatment of foreign digital service invoices, and the opportunity cost of capital tied up in domestic hardware versus preserved for core business investment.

7.3 Financial Friction and Foreign Exchange Billing

As detailed in Section 4.3, the mechanics of paying a foreign cloud vendor from Nepal have historically required a prepaid dollar card subject to an annual ceiling, together with tax deducted at source obligations and value added tax considerations on imported digital services. The 2026 amendment expanding IT sector specific allowances materially eases this constraint for registered technology businesses and foreign currency earners, but general consumers and unregistered small businesses remain subject to more restrictive limits, meaning that the

practical foreign exchange burden of hyperscaler adoption in Nepal continues to vary significantly by the legal and tax registration status of the purchasing entity rather than by technical need alone.

7.4 Feature Parity and Ecosystem Maturity

International hyperscalers offer a substantially broader and more mature portfolio of managed services than any domestic Nepali provider currently matches, including managed relational and NoSQL databases, managed Kubernetes services, serverless function platforms, content delivery networks with global points of presence, and increasingly accessible artificial intelligence and machine learning pipelines, including managed large language model inference endpoints. Domestic providers, by contrast, presently concentrate on colocation, virtual private servers, and basic infrastructure as a service offerings, without the managed platform layer that allows international competitors to reduce a client's operational engineering burden. This ecosystem gap is the single most significant reason that Nepal's technology export sector, which must remain internationally competitive on engineering velocity, continues to rely overwhelmingly on foreign hyperscalers despite the data sovereignty and latency advantages of domestic alternatives.

Table 6: Comparative Benchmark of Domestic Cloud and Colocation Providers versus International Hyperscalers

Criterion	**Domestic Providers (e.g. Ncell, Nepal Telecom, local colocation)**	**International Hyperscalers (AWS, Azure, Google Cloud)**
Data Sovereignty	High; data remains within national jurisdiction and subject to Directive 2081 oversight	Low to moderate; data typically resides in India or Singapore regions, subject to foreign jurisdiction
Latency for Domestic Users (indicative)	Low, generally single digit to low double digit milliseconds within Kathmandu Valley	Moderate, generally in the tens of milliseconds range to nearest regional data center
Managed Platform Services (PaaS, serverless, managed AI/ML)	Limited to largely unavailable at present	Extensive and continuously expanding
Billing Currency and Foreign Exchange Exposure	Predominantly Nepali rupee; minimal foreign exchange exposure	United States dollar denominated; subject to NRB card limits and currency risk
Elasticity and Auto Scaling	Limited; capacity often provisioned manually	Extensive, on demand elastic scaling
Regulatory Compliance Burden	Directive 2081 registration; simpler for purely domestic tenants	Requires tenant level compliance mapping to Nepali law despite provider being foreign incorporated
Disaster Recovery Geographic Redundancy	Limited primarily to Kathmandu Valley and Hetauda	Global, multi region redundancy available

Table 7: Indicative Total Cost of Ownership Multi Year Model for Nepali Enterprises

Cost Component	**Domestic Private/Colocation Model (Illustrative, 3 Year Horizon)**	**Public Hyperscaler Model (Illustrative, 3 Year Horizon)**
Upfront Capital Investment	High; hardware procurement, import duties, and installation	Minimal; no hardware ownership required
Ongoing Operational Cost Pattern	Relatively stable; power, cooling, staffing, and maintenance in Nepali rupees	Variable and usage sensitive; billed in United States dollars
Foreign Exchange Exposure	Low, confined largely to initial hardware import	High, recurring monthly settlement subject to NRB card and allowance limits
Scalability Cost Behavior	Step function; capacity added in discrete hardware increments	Continuous; capacity added incrementally with usage based billing
Disaster Recovery Cost	Requires separate secondary site investment (e.g. Hetauda style facility)	Included or readily available as a managed, multi region service
Total Cost Sensitivity	Sensitive to import costs, power reliability, and local technical staffing availability	Sensitive to currency depreciation, tax treatment, and evolving NRB foreign exchange allowances

CHAPTER 8: RISKS, CYBERSECURITY, AND OPERATIONAL CHALLENGES

8.1 Data Sovereignty, Extraterritorial Jurisdiction, and Vendor Lock In

Reliance on foreign hyperscalers exposes Nepali organizations, particularly those handling regulated financial or personal data, to extraterritorial legal jurisdiction over their data, since infrastructure physically located in India, Singapore, or elsewhere is governed in the first instance by the law of that jurisdiction, potentially in tension with obligations under the Individual Privacy Act 2075 and sector specific Nepali regulation. Vendor lock in is a related concern: once an organization has built its systems around a specific hyperscaler's proprietary managed services, migration to a domestic or alternative foreign provider becomes technically costly and organizationally disruptive, reducing Nepal's practical bargaining position and long term data sovereignty over time.

8.2 Cyber Threat Landscape

Cloud hosted Nepali systems, whether domestic or foreign, face the same generic cloud threat categories documented globally: distributed denial of service (DDoS) attacks capable of disrupting public facing government and financial services; misconfigured cloud object storage that inadvertently exposes sensitive data to the public internet; credential leaks arising from weak identity and access management practices; and ransomware capable of encrypting production systems and demanding payment for restoration. Nepal's relatively young formal cyber security governance architecture, embodied in the National Cyber Security Policy 2080, explicitly acknowledges that the country's institutional capacity to detect and respond to these threats remains under development.

8.3 Public Sector Institutional Bottlenecks

Several institutional bottlenecks constrain the Government Integrated Data Center's cyber security posture relative to global best practice. Nepal's public sector currently lacks a dedicated, continuously staffed Security Operations Center (SOC) capable of around the clock monitoring and incident response for government systems, and the civil service faces a documented shortage of personnel holding internationally recognized cloud security certifications. Compounding this, the Public Procurement Act 2063 has historically structured government technology procurement around lowest bidder evaluation criteria, which can favor minimally compliant vendors over those offering higher reliability, better documented security architecture, or stronger service level guarantees, a structural tension that policy makers must address if government cloud security posture is to improve materially.

8.4 International Network Transit Vulnerability

As a landlocked nation, Nepal has no direct access to submarine fiber optic cable landing stations and depends entirely on terrestrial optical fiber links transiting India, and to a lesser degree China, for its international internet and cloud connectivity. This dependency creates both a latency penalty for any workload hosted outside Nepal and a structural single point of failure risk: disruption to cross border transit links, whether from physical damage, political disruption, or capacity constraints during peak demand, can degrade or sever Nepal's connectivity to foreign hyperscaler regions even when the hyperscaler's own infrastructure is functioning normally. The September 2025 unrest illustrated a related but distinct risk, demonstrating that even purely domestic infrastructure threats, such as fire near a data center facility, can jeopardize continuity of government digital services regardless of international connectivity status, reinforcing the case for continued investment in geographically distributed domestic disaster recovery capacity alongside international transit diversification.

Table 8: Risk Categorization, Operational Impact, and Mitigation Matrix for Cloud Deployments in Nepal

Risk Category	Operational Impact	Recommended Mitigation
Data sovereignty and extraterritorial jurisdiction	Regulated data exposed to foreign legal process; compliance ambiguity under Individual Privacy Act 2075	Data classification policies; domestic hosting for regulated data categories; contractual data residency clauses
Vendor lock in	Reduced negotiating leverage; high migration cost over time	Adoption of open standards and containerized, portable architectures; periodic multi cloud cost review
DDoS and public facing service disruption	Downtime for citizen facing and financial services	Content delivery network and DDoS mitigation services; traffic scrubbing partnerships
Misconfiguration and credential leakage	Unauthorized data exposure	Automated configuration auditing; mandatory multi factor authentication and least privilege access control
Absence of around the clock SOC in government	Delayed detection and response to incidents	Establishment of a dedicated national government SOC under NITC or the National Cyber Security Center
Shortage of certified cloud security personnel in civil service	Weak in house incident response capacity	Structured certification programs (AWS, Azure, Certified Kubernetes Administrator) for civil servants
Lowest bidder procurement bias	Selection of minimally compliant infrastructure vendors	Procurement reform incorporating security and reliability weighted evaluation criteria
Cross border transit dependency	Latency and connectivity risk from India/China transit reliance	Transit route diversification; continued domestic disaster recovery investment
Concentration of critical services in limited physical sites	Systemic risk from localized events (e.g. September 2025 unrest)	Geographic expansion of data centers (e.g. Khumaltar, Nepalgunj, provincial sites)

CHAPTER 9: EMERGING TRENDS AND FUTURE OUTLOOK

Several emerging trends are likely to shape the next phase of Nepal's cloud adoption trajectory. First, there is a credible prospect of sovereign cloud and local availability zone development, whether through a regional hyperscaler establishing a local edge point of presence or local zone in Kathmandu, or through a domestically operated provider such as Ncell or Nepal Telecom expanding its listed cloud services toward genuine self service, elastically scaling offerings under the Directive 2081 licensing framework. Such a development would materially reduce the latency and data sovereignty tradeoffs described in Chapter 7.

Second, artificial intelligence and graphics processing unit (GPU) cloud demand is likely to grow as Nepali enterprises and researchers seek domestic computing capacity for large language model fine tuning in Devanagari script and for predictive analytics tailored to local languages and use cases, a demand that current domestic infrastructure is not yet well positioned to serve and that may, at least in the near term, continue to be met through foreign hyperscaler GPU instances.

Third, Nepal's federal governance structure creates a plausible pathway toward hybrid mesh architectures and cloud native federalism, in which edge computing clusters are distributed across the seven provincial governments, consistent with NITC's own stated intent to expand government data center capacity beyond Kathmandu Valley to sites such as Khumaltar, Nepalgunj, and eventually all seven provinces. Such a distributed model would reduce the systemic concentration risk highlighted by the September 2025 stress test while bringing government computing capacity closer to provincial and local level service delivery points.

CHAPTER 10: CONCLUSION AND STRATEGIC SYNTHESIS

Nepal's cloud computing adoption trajectory is best characterized as an emergent, hybrid, and rapidly formalizing landscape rather than either a mature public cloud market or a purely traditional on premises environment. The public sector's Government Integrated Data Center and its Hetauda disaster recovery counterpart have proven their value under genuine stress, most visibly during the September 2025 unrest, yet remain constrained by conventional virtualization architecture, incomplete backup coverage, and concentration in a small number of physical sites. The private sector exhibits a pronounced bifurcation between a cautious, domestically anchored BFSI sector operating under Nepal Rastra Bank's guidelines, and a globally integrated technology export sector whose competitiveness depends on continued, largely unconstrained access to international hyperscaler platforms.

The enactment of the Data Center and Cloud Service (Operation and Management) Directive 2081 marks a genuine inflection point, introducing for the first time a unified registration and oversight regime spanning government, public, and private data centers and cloud service providers, and Ncell's early listing demonstrates that domestic providers can meet its operational bar. At the same time, the Directive's two year review horizon and the still evolving foreign exchange allowances for the information technology sector indicate that Nepal's regulatory architecture for cloud computing remains a work in progress rather than a settled framework. Bridging the gap between strict local regulatory mandates, including data sovereignty and security obligations, and the agile, elastic operational demands of modern digital services will require sustained, coordinated action across government, the central bank, infrastructure operators, private enterprise, and academia, the specific contours of which are set out in the recommendations that follow.

CHAPTER 11: ACTIONABLE RECOMMENDATIONS

11.1 Government Policy Makers and the Ministry of Communication and Information Technology

- Complete the scheduled two year review of Directive 2081 with published, transparent findings, and use the review to close ambiguities regarding cross border data transfer for non regulated data categories.
- Reform Public Procurement Act 2063 evaluation criteria for information technology and data center procurement to weight security architecture, reliability, and total cost of ownership alongside price, rather than continuing to default to lowest bidder selection.
- Accelerate the provincial data center expansion already indicated by NITC, prioritizing geographic diversification to reduce the single site concentration risk exposed by the September 2025 unrest.

11.2 Nepal Rastra Bank and Financial Regulators

- Formalize and publish clear, durable guidance on the 2026 information technology sector foreign exchange allowances, since the sector currently relies on secondary reporting rather than a single authoritative published circular.
- Issue updated, cloud specific guidance for BFSI institutions clarifying which data categories may be hosted on foreign public cloud infrastructure under appropriate encryption and contractual safeguards, reducing reliance on an overly conservative blanket interpretation of the 2012 IT Guidelines.

11.3 Public Cloud Infrastructure Operators (NITC and the Integrated Data Management Center)

- Invest in container orchestration and automated horizontal scaling capability to convert the Government Integrated Data Center from a shared virtualization facility into a genuine self service internal cloud platform.
- Establish comprehensive, mandatory backup coverage for all hosted government agencies, closing the gap acknowledged following the 2023 data loss incident.
- Establish a dedicated, continuously staffed Security Operations Center for government systems, potentially in coordination with the National Cyber Security Center envisioned under the National Cyber Security Policy 2080.

11.4 Private Sector and Commercial Data Center Operators

- Pursue internationally recognized tier certification and transparent publication of technical specifications to build enterprise and government client confidence, following Ncell's precedent as the first Directive 2081 listed provider.
- Invest in developing managed platform as a service offerings, including managed databases and container orchestration, to narrow the feature parity gap with international hyperscalers documented in Chapter 7.

- Adopt transparent, published pricing to enable Nepali enterprises to perform accurate total cost of ownership comparisons against foreign hyperscaler alternatives.

11.5 Academia and Human Capital Development

- Integrate cloud architecture, containerization, and cloud security curricula into Bachelor of Science in Computer Science and Information Technology and equivalent programs, building on the analytical framework developed in this report.
- Encourage and subsidize internationally recognized certifications, including Amazon Web Services, Microsoft Azure, Google Cloud, and Certified Kubernetes Administrator credentials, to address the shortage of certified cloud professionals documented in Chapter 8.
- Establish structured academia government partnerships, potentially through internship placements at NITC and the Department of Information Technology, to channel emerging talent directly into public sector cloud modernization efforts.

Table 9: Stakeholder Action Matrix with Prioritized Implementation Horizons

Stakeholder	Priority Action	Implementation Horizon
MoCIT / Government Policy Makers	Publish Directive 2081 two year review findings; reform procurement evaluation criteria	Short to medium term (1 to 2 years)
Nepal Rastra Bank	Formalize IT sector foreign exchange allowance guidance; issue cloud specific BFSI hosting guidance	Short term (within 1 year)
NITC / Integrated Data Management Center	Deploy container orchestration and automated scaling; establish government SOC; complete backup coverage	Medium term (2 to 4 years)
Commercial Data Center and Cloud Providers	Pursue tier certification; develop managed PaaS offerings; publish transparent pricing	Medium term (2 to 4 years)
Academic Institutions (e.g. Himalaya College of Engineering)	Integrate cloud curricula; expand certification support; formalize government internship pipelines	Ongoing, beginning immediately

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